

IVF Clinic Management System

Lab Technician Role

User Stories, Use Cases & User Flows

Software Engineering – Laboratory Deliverable | 18 March 2026 | Silvja Fejzulla

1. Lab Technician — Role Overview

The Lab Technician is a core clinical role in the IVF Clinic Management System. Lab Technicians are responsible for all laboratory-side operations: processing patient samples, conducting and reporting tests, recording embryo development, maintaining the donation bank, and ensuring equipment and quality-control compliance. Their work directly informs clinical decisions made by doctors and determines treatment outcomes for patients.

Core Responsibilities :

Sample Processing: Receive, label, and track patient and donor samples from arrival through to analysis, maintaining full chain-of-custody at every stage.

Test Reporting: Process lab test orders requested by doctors, upload completed test reports linked to the patient record, and flag any abnormal results automatically for urgent clinical review.

Embryology Support: Record daily embryo development observations throughout the IVF culture period, document cryopreservation details for frozen embryo transfer planning, and review doctor instructions for each embryo.

Donation Bank Management: Maintain accurate sperm, egg, and embryo donation stock levels, update donor sample screening statuses, and ensure expired or rejected samples are removed from the assignable pool.

Equipment & Quality Control: Keep lab equipment calibration and maintenance records up to date, perform and document QC checks on instruments and reagents, and escalate any compliance failures before patient testing resumes.

Audit & Compliance: Log all custody events and record changes in a timestamped, immutable audit trail to satisfy regulatory inspection requirements.

Attribute	Detail
Role	Lab Technician
System Modules Accessed	Test Reports, Embryology, Donation Bank, Equipment Management, Sample Tracking, Quality Control
Interacts With	Doctor (receives test requests, shares results), Hospital Administrator (low-stock alerts), Patient (indirect via test results)
Key Constraints	Can only access records for assigned patients/cycles; all entries are timestamped and immutable once confirmed; must authenticate for all inventory changes; must comply with clinical and regulatory traceability requirements.

2. Refined & New User Stories

The stories below represent the full scope of Lab Technician functionality. Stories marked 'Refined' are improved versions of the original requirements. Stories marked 'New' are additions identified during role analysis. Priorities: High = must-have for MVP; Medium = important but deferrable; Low = nice-to-have.

2a. Test Request & Reporting

ID	User Story	Priority	Status	Refinement Notes
US_LT_01	As a lab technician, I want to view the queue of pending lab test requests ordered by urgency and due time so that I can prioritise and process the most critical samples first.	High	New	Addresses workload visibility; prevents urgent samples from being missed in high-volume sessions.
US_LT_02	As a lab technician, I want to receive and label patient samples upon arrival by scanning a barcode or entering a sample ID so that each specimen is uniquely identified and tracked from collection through to analysis.	High	New	Closes the gap between sample collection and test processing; enables full chain-of-custody.
US_LT_03	As a lab technician, I want to upload completed test reports linked to the patient record and the requesting doctor's order so that results are immediately available for clinical review.	High	Refined	Original (US_25) lacked the explicit link to the doctor's request. Added 'order linkage' for traceability.
US_LT_04	As a lab technician, I want the system to automatically flag abnormal test results based on predefined clinical reference ranges and send an urgent alert to the requesting doctor so that critical findings are never missed.	High	Refined	Original (US_29) did not specify how flagging is triggered. Added 'predefined reference ranges' to make acceptance criteria testable.

2b. Embryology & IVF Cycle Support

ID	User Story	Priority	Status	Refinement Notes
US_LT_05	As a lab technician, I want to record daily embryo development observations — including cell count, fragmentation percentage, morphology score, and free-text notes — so that the medical team can accurately assess embryo quality throughout the culture period.	High	Refined	Original (US_26) listed only some observation fields. Expanded to include fragmentation percentage and free-text notes for completeness.

ID	User Story	Priority	Status	Refinement Notes
US_LT_06	As a lab technician, I want to record embryo cryopreservation details (storage tank, cane, straw position, freezing date, and vitrification method) so that frozen embryos can be precisely located and retrieved for future frozen embryo transfer (FET) cycles.	High	New	Cryopreservation is a critical IVF sub-process not covered by existing stories. Directly supports FET planning.
US_LT_07	As a lab technician, I want to view any clinical notes or transfer/freeze instructions added by the doctor to an embryo record so that I carry out the correct procedure for each embryo.	Medium	New	Closes the communication loop between doctor and lab; reduces procedural errors.

2c. Donation Bank Management

ID	User Story	Priority	Status	Refinement Notes
US_LT_08	As a lab technician, I want to update donation bank stock levels (sperm, egg, and embryo quantities, collection dates, and storage locations) so that sample availability is always accurate and the system reflects the physical inventory.	High	Refined	Original (US_27) merged donor status and stock quantity into one story. Split into separate concerns for clarity.
US_LT_09	As a lab technician, I want to update a donor's sample screening status (Pending, Cleared, Rejected) so that only screened and eligible donor samples are available for use in patient treatments.	High	New	Donor sample screening is a regulatory requirement; keeping it separate from stock quantity improves traceability.

2d. Sample Tracking & Chain-of-Custody

ID	User Story	Priority	Status	Refinement Notes
US_LT_10	As a lab technician, I want to log every sample chain-of-custody event — including received, processed, stored, transferred, and discarded — with a timestamp and my technician ID so that full traceability is maintained for regulatory inspections and audits.	Medium	Refined	Original (US_30) stated the goal but not the required log fields. Added event types and technician ID for testable acceptance criteria.

2e. Equipment Management & Quality Control

ID	User Story	Priority	Status	Refinement Notes
US_LT_11	As a lab technician, I want to view and update lab equipment records (instrument name, last calibration date, next calibration due date, and maintenance status) so that all instruments remain compliant with clinical standards.	Medium	Refined	Original (US_28) was vague about record fields. Added specific fields to make acceptance criteria measurable.
US_LT_12	As a lab technician, I want to schedule and record preventive maintenance tasks for lab equipment, including the date performed and technician responsible, so that instrument downtime and compliance failures are minimised.	Medium	New	Complements equipment record-keeping with proactive scheduling; reduces risk of unplanned equipment failure.
US_LT_13	As a lab technician, I want to perform and record quality control (QC) checks for reagents and instruments, including the QC result (pass/fail) and any corrective action taken, so that test accuracy and reliability are continuously assured.	High	New	QC documentation is mandatory under clinical laboratory accreditation standards (e.g. ISO 15189).
US_LT_14	As a lab technician, I want to receive a low-stock alert when critical lab supplies or reagents fall below a configurable minimum threshold so that I can initiate a restocking request before operations are disrupted.	Medium	New	Prevents operational disruption; threshold value should be configurable by the lab manager.
US_LT_15	As a lab technician, I want to generate a daily lab activity summary report showing tests processed, tests pending, results flagged, and equipment issues so that the lab supervisor has real-time visibility into throughput and workload.	Low	New	Supports lab management oversight; useful for shift handover and performance monitoring.

3. Use Cases — User Flows, Restrictions & Constraints

Each use case describes the complete user flow for a key Lab Technician action, including the main success scenario, alternative and exception paths, postconditions, and any relevant restrictions or constraints. The format follows the structure used in the course case study.

3.1 UC_LT_01: Upload Test Report

Use Case Name	Upload Test Report
ID	UC_LT_01
Actors	Lab Technician (primary), Doctor (notified), Patient (indirect)

Preconditions	• Lab Technician is authenticated and logged in. • A lab test order exists in the system (created by a doctor). • The test has been performed and results are ready to upload.
Main Success Flow	1. Lab Technician navigates to the Test Queue and views pending orders. 2. Lab Technician selects the relevant test order (filtered by patient or order ID). 3. System displays the order details: patient name, test type, requesting doctor, and any clinical notes. 4. Lab Technician uploads the result file (PDF or structured data) or enters result values manually. 5. Lab Technician reviews the result before submission. 6. System compares result values against predefined reference ranges. 7. System saves the report linked to the patient record and the doctor's order. 8. System timestamps the submission and records the technician's ID. 9. System notifies the requesting doctor that results are ready. 10. Doctor reviews and approves visibility to patient; patient can then view results in their portal.
Alternative Flows	4a. Result entry error: System highlights invalid or out-of-range values; technician must confirm or correct before submitting. 6a. Abnormal result detected: System auto-flags the result with a colour-coded indicator and sends an urgent alert to the doctor (see US_LT_04). 2a. Order not found: Technician can search by patient ID or doctor name; if still not found, they contact the requesting doctor.
Postconditions	• Test report is stored and linked to the patient record. • Requesting doctor receives a notification. • If flagged, an urgent alert is sent immediately. • Audit log records the upload: technician ID, timestamp, and order reference.
Constraints & Restrictions	• Only authenticated lab technicians can upload results. • Results must be linked to an existing doctor-created order. • Uploaded files must not exceed 25 MB; accepted formats: PDF, JPEG, PNG, CSV. • Results cannot be deleted once submitted; corrections require an amended report with a reason. • Patient access to results is gated by doctor review (configurable per test type). • All uploads are logged for regulatory audit purposes.

3.2 UC_LT_02: Record Embryo Development

Use Case Name	Record Embryo Development & Cryopreservation
ID	UC_LT_02
Actors	Lab Technician (primary), Doctor (views and adds instructions), Patient (read-only view)
Preconditions	• Lab Technician is logged in. • An active IVF cycle exists with fertilised embryos recorded in the system. • The lab technician is assigned to this patient's cycle.

Main Success Flow	11. Lab Technician navigates to the Embryology Module. 12. Lab Technician searches for and selects the patient's active IVF cycle. 13. System displays the embryo list with the most recent development data and any doctor instructions. 14. Lab Technician selects an embryo to update. 15. Lab Technician enters the day's observation data: development day, cell count, fragmentation percentage, morphology score (e.g. Gardner grade), and free-text observation notes. 16. Lab Technician optionally uploads a microscopy image (max 10 MB). 17. System validates mandatory fields (day, cell count, grade). 18. System saves the entry as immutable, with a timestamp and technician ID. 19. System updates the embryo's status on the doctor's and patient's dashboards. 20. If the doctor has added a freeze or transfer instruction, system highlights it prominently. 21. For cryopreservation: Lab Technician selects 'Freeze Embryo' and enters storage location (tank, cane, straw), freezing date, and vitrification method. 22. System records the cryopreservation event and links it to the embryo record.
Alternative Flows	3a. No embryos recorded yet: System prompts technician to record initial fertilisation results (day 0 data). 5a. Data value outside expected range: System warns the technician but allows entry with a mandatory confirmation note explaining the deviation. 7a. Mandatory field missing: System blocks submission and highlights the missing field. 6a. Cycle cancelled by doctor: System displays a cancellation notice; technician can add a final observation but cannot transition the cycle.
Postconditions	• Embryo development record is updated; entry is immutable once confirmed. • Doctor and patient views reflect the new data in near-real-time. • If cryopreserved, storage location is recorded and the embryo status is updated to 'Frozen'. • All entries carry a timestamp and technician ID.
Constraints & Restrictions	• Only assigned lab technicians can record or update embryo data for a cycle. • Entries are append-only; once confirmed they cannot be edited — corrections must be logged as a new entry with a reference to the original. • Minimum mandatory fields: development day, cell count, and morphology grade. • Microscopy images must not exceed 10 MB per upload. • Cycle phase transitions (e.g. Fertilisation → Embryo Culture) require doctor authorisation. • Cryopreservation records must include storage tank, cane, straw position, date, and method. • Patient data is subject to data-protection regulations; access is role-restricted.

3.3 UC_LT_03: Manage Donation Bank Stock

Use Case Name	Manage Donation Bank Stock & Donor Sample Status
ID	UC_LT_03
Actors	Lab Technician (primary), Hospital Administrator (receives low-stock alerts)
Preconditions	• Lab Technician is logged in. • Donation Bank module is configured with at least one donor record.

Main Success Flow	23. Lab Technician navigates to the Donation Bank module. 24. System displays the current inventory: sample type (sperm/egg/embryo), donor ID, quantity, collection date, expiry date, storage location, and screening status. 25. Lab Technician selects an action: Add New Sample, Update Quantity, Update Screening Status, or Mark as Expired. 26. Add New Sample: Technician enters donor ID, sample type, quantity, collection date, storage location, and initial screening status (Pending). 27. System validates the donor ID against registered donor records. 28. System adds the sample to inventory and logs a chain-of-custody 'Received' event. 29. Update Screening Status: Technician sets status to Cleared or Rejected with a mandatory reason note. 30. System updates the sample record; Rejected samples are immediately removed from the assignable pool. 31. System sends a low-stock alert to the administrator if any sample type falls below the configured minimum threshold. 32. System displays the updated inventory view.
Alternative Flows	4a. Donor ID not found: System prompts the technician to register the donor before adding a sample. 9a. Quantity falls below minimum threshold: System immediately triggers a low-stock notification to the administrator. Expiry path: System automatically flags samples whose expiry date is within 30 days; expired samples are locked from patient assignment.
Postconditions	• Donation bank inventory is updated and reflects physical stock. • All changes are logged as chain-of-custody events with technician ID and timestamp. • Rejected or expired samples are removed from the assignable pool. • Low-stock alert is sent to the administrator if thresholds are breached.
Constraints & Restrictions	• All donor samples must be linked to a registered and uniquely identified donor. • Expired or rejected samples cannot be assigned to patients. • Every inventory change requires technician authentication and is permanently logged. • Storage location must be recorded for every sample (physical traceability). • Screening status changes require a mandatory reason note. • Minimum stock thresholds are configurable by the lab manager or administrator.

3.4 UC_LT_04: Log Sample Chain-of-Custody

Use Case Name	Log Sample Chain-of-Custody Event
ID	UC_LT_04
Actors	Lab Technician (primary)
Preconditions	• Lab Technician is logged in. • A patient sample or donor sample record exists in the system.

Main Success Flow	33. Lab Technician navigates to the Sample Tracking module. 34. Lab Technician scans the sample barcode or manually enters the sample ID. 35. System retrieves and displays the sample's current custody history. 36. Lab Technician selects the event type: Received, Processed, Stored, Transferred, or Discarded. 37. For 'Transferred': Technician enters the destination (person or location receiving the sample). 38. For 'Discarded': Technician enters the disposal reason and method. 39. System records the event with: event type, timestamp, technician ID, and any additional notes. 40. System displays the updated chain-of-custody log for the sample.
Alternative Flows	2a. Sample ID not found: System returns an error; technician must verify the sample ID or register the sample first. 4a. Discarded event: System requires a mandatory reason code and disposal confirmation before saving.
Postconditions	• A new chain-of-custody entry is created and linked to the sample record. • The entry is immutable; corrections require a new log entry with a reference and explanation. • The full custody history is available for regulatory audit at any time.
Constraints & Restrictions	• All custody events must be linked to an authenticated technician. • Event records are immutable once created. • Disposal events require a mandatory reason code selected from a controlled list. • Transfer events must record the recipient identity (name and role or department). • Chain-of-custody logs must be retained for a minimum period in line with local regulatory requirements.

3.5 UC_LT_05: Manage Lab Equipment & Quality Control

Use Case Name	Manage Lab Equipment Records & Perform Quality Control Checks
ID	UC_LT_05
Actors	Lab Technician (primary)
Preconditions	• Lab Technician is logged in. • Lab equipment records are registered in the Equipment Management module. • QC reference ranges / pass-fail criteria are configured in the system.

Main Success Flow	41. Lab Technician navigates to the Equipment Management module. 42. System displays the equipment list with: instrument name, last calibration date, next calibration due date, and current status (Active, Maintenance Required, Out of Service). 43. Lab Technician selects an instrument to update. 44. Lab Technician updates the calibration date or maintenance status and adds a note. 45. System saves the update with a timestamp and technician ID. 46. System highlights instruments whose next calibration date is approaching (within 7 days) or overdue. 47. For QC check: Lab Technician selects 'Record QC Check' for the relevant instrument or reagent lot. 48. Lab Technician enters QC result values and selects Pass or Fail. 49. If Fail: Lab Technician enters the corrective action taken (e.g. reagent replaced, instrument recalibrated). 50. System saves the QC record with timestamp, technician ID, and corrective action if applicable. 51. System displays a warning if an instrument has failed QC and has not yet been cleared; results from that instrument are flagged as 'Pending QC clearance'.
Alternative Flows	6a. Calibration overdue: System generates a maintenance alert and changes the instrument status to 'Calibration Required'; the instrument cannot be used for patient tests until recalibrated. 8a. QC Fail: System blocks the use of the failed instrument or reagent lot for patient samples until a Pass QC record is entered. 3a. Equipment record not found: Technician can create a new equipment record by entering instrument name, serial number, manufacturer, and initial calibration date.
Postconditions	• Equipment record is updated with the latest calibration date, status, and any QC results. • QC records are stored with timestamp and technician ID. • Instruments with failed QC or overdue calibration are flagged and blocked from patient-test use. • Corrective actions are documented for audit purposes.
Constraints & Restrictions	• Only authenticated lab technicians can update equipment or QC records. • Instruments with overdue calibration or failed QC cannot be used for patient testing until cleared. • QC checks must reference a specific reagent lot number or instrument serial number. • All records (calibration and QC) are immutable once saved; amendments must reference the original entry. • Calibration intervals and QC pass/fail thresholds are set by the lab manager and cannot be modified by a regular technician. • Equipment and QC records must be retained for a minimum period per clinical laboratory accreditation standards (e.g. ISO 15189).

4. Summary of Changes & Additions

The table below summarises all refinements made to the original Lab Technician user stories and identifies the new stories added through this analysis.

Story ID	Story Title	Status	Key Change / Rationale
US_LT_01	View pending test request queue	New	Provides workload visibility; prevents critical samples being missed.
US_LT_02	Receive and label patient samples on arrival	New	Closes the gap between sample collection and processing; enables full chain-of-custody from entry.
US_LT_03	Upload completed test reports (was US_25)	Refined	Added explicit linkage to doctor's order for traceability; clarified file and data entry options.
US_LT_04	Auto-flag abnormal results (was US_29)	Refined	Specified triggering mechanism (reference ranges) to make acceptance criteria testable.
US_LT_05	Record embryo development observations (was US_26)	Refined	Expanded observation fields to include fragmentation % and free-text notes.
US_LT_06	Record embryo cryopreservation details	New	Covers a critical IVF sub-process (FET planning) absent from original stories.
US_LT_07	View doctor's embryo instructions	New	Closes doctor-lab communication loop; reduces procedural errors.
US_LT_08	Update donation bank stock inventory (was US_27 part)	Refined	Split US_27 into stock quantity and donor screening for clarity.
US_LT_09	Update donor sample screening status	New	Regulatory requirement; separated from stock management for distinct traceability.
US_LT_10	Log sample chain-of-custody events (was US_30)	Refined	Added required log fields (event types, technician ID) for testable acceptance criteria.
US_LT_11	View and update equipment records (was US_28)	Refined	Made record fields explicit (calibration dates, status) for measurable acceptance criteria.
US_LT_12	Schedule and record preventive maintenance	New	Proactive maintenance scheduling to reduce unplanned downtime.
US_LT_13	Perform and record QC checks	New	ISO 15189 requirement; ensures test accuracy and reagent traceability.
US_LT_14	Receive low-stock alerts for lab supplies	New	Prevents operational disruption; threshold is configurable.
US_LT_15	Generate daily lab activity summary report	New	Supports supervisor oversight and shift handover.

End of Lab Technician Deliverable | Silvja Fejzulla | March 2026